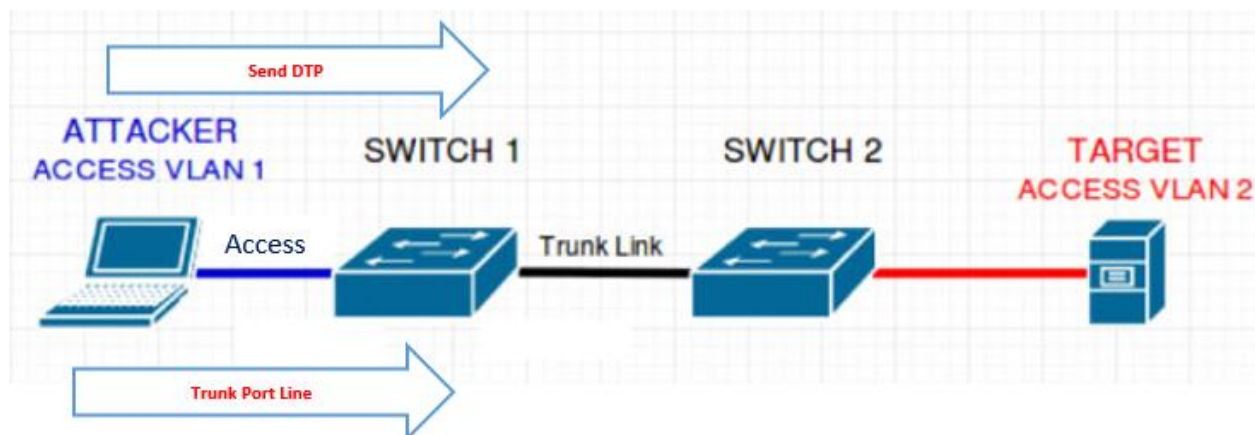
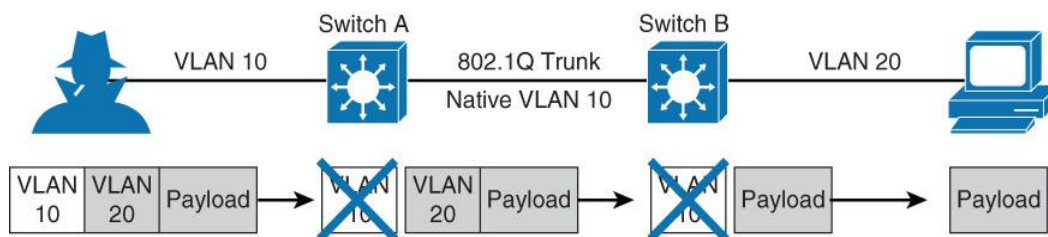
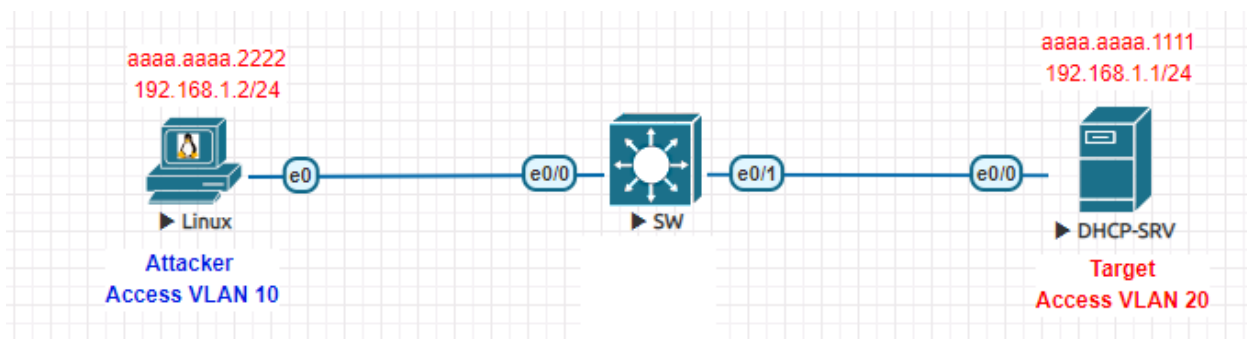


VLAN Hopping Attack:

- o VLAN hopping is security exploit method of attacking networked resources on VLAN.
- o VLAN Hopping is attack where attacker able to send traffic from one VLAN into another.
- o Attacks when an attacker tricks a switch into allowing traffic to hop to a different VLAN.
- o A VLAN hopping attack allows a frames from one VLAN to pass into another VLAN.
- o Attacker connects to VLAN to gain access to traffic on other Virtual Local Area Network.
- o Other VLAN (Virtual Local Area Network) would normally not be accessible.
- o There are two VLAN hopping exploit methods **Switch Spoofing** and **Double Tagging**.
- o Attacker send spoofed trunk negotiation and the interface configured as a trunked.
- o The second Virtual Local Area Network hopping method is called “Double-Tagging”
- o By using this native VLAN and additional VLAN tag, can hop from one VLAN to another.
- o Perform double tagging, to hop through multiple Cisco Switches in LAN Network.
- o With double tagging, the attacker is now able to hop to another Virtual LAN.
- o Disable any Automated Trunk Negotiation Dynamic Trunking Protocol on Switches.
- o To prevent double tagging, change the native VLAN of the Cisco Switch trunk ports.

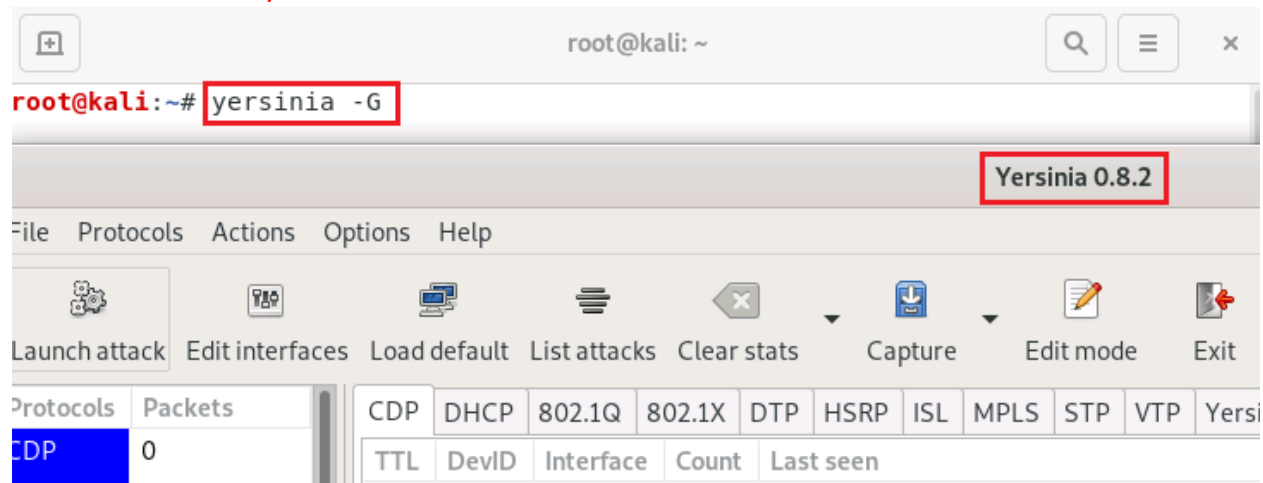




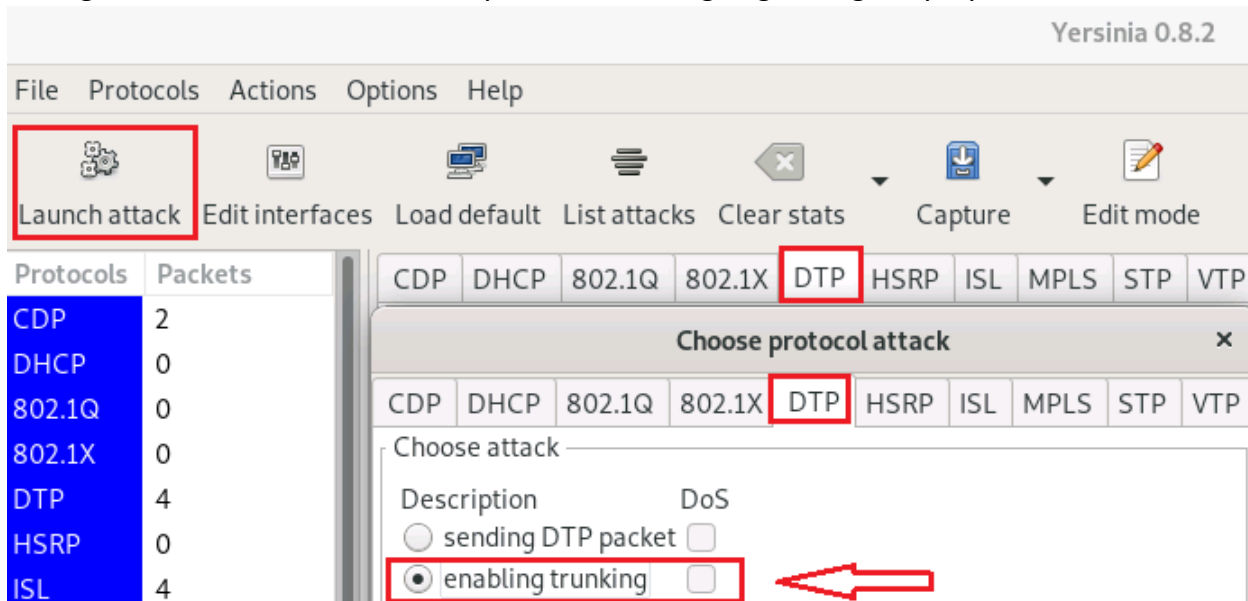
Switch Basic Configuration

```
SW(config)#vlan 10
SW(config-vlan)#name HR
SW(config-vlan)#vlan 20
SW(config-vlan)#name IT
SW(config)#interface e0/0
SW(config-if)#switchport access vlan 10
SW(config)#interface e0/1
SW(config-if)#switchport access vlan 20
SW# show interfaces trunk
There is no trunk available
```

To launch Yersinia: **yersinia -G**



Now to send a DTP message click "Launch attack", click the tab "DTP", click "enable trunking" and last click "OK". Yersinia will then send out a DTP message and within a few seconds, a trunking link will be established. , the attacker will then have access to all traffic flowing through other VLANs and can directly attack without going through any layer 3 devices.



Now check again there is Trunk created automatically now the attacker will then have access to all VLAN in this case VLAN 1, 10 and 30.

SW# **show interfaces trunk**

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	auto	n-802.1q	trunking	1
Port Et0/0	vlangs allowed on trunk			
	1-4094			
Port Et0/0	vlangs allowed and active in management domain			
	1,10,20			
Port Et0/0	vlangs in spanning tree forwarding state and not pruned			
	none			

Switch Spoofing:

- o To prevent switch spoofing attack, disable Dynamic Trunking Protocol (DTP) on ports.
- o VLAN Hopping Attack be mitigated by configuring end users switch ports in access mode.
- o Hardcode all the access ports of Cisco Switches as access port and disable DTP everywhere.
- o Shutdown all the interfaces in you every Cisco Switches in network, which are not in use.
- o Never leave an access port of Cisco Switches in "dynamic desirable", "dynamic auto" mode.

Switch Configuration
SW(config)#interface ethernet0/0 SW(config-if)#switchport mode access SW(config-if)#switchport nonegotiate
SW(config)#interface range ethernet0/0 -3 SW(config-if)#switchport access vlan 88 SW(config-if)# shutdown

Native VLAN:

- o Native VLAN is the default Virtual Local Area Network that all untagged traffic goes into.
- o By default, in every Cisco Switch Native VLAN is Virtual Local Area Network (VLAN) 1.
- o For the purpose of security, the native Virtual Local Area Network must be changed.
- o Restrict VLANs on trunk ports also disable Dynamic Trunking Protocol on Cisco Switches.
- o To prevent a Double Tagging attack, keep native VLAN of all trunk ports different from user.
- o To prevent double tagging be sure that native VLAN on every trunk port is an unused VLAN.

Implications of Native VLAN
SW# show interfaces trunk
SW(config)#interface ethernet0/3 SW(config-if)#switchport trunk native vlan 99 SW(config-if)#switchport trunk allowed vlan 10, 20